

Calculate the gradient of the following function:

$$f : \mathbb{R}^3 \rightarrow \mathbb{R} : f(x_1, x_2, x_3) = e^{x_1} + x_1^2 x_3 - x_1 x_2 x_3.$$

What is the directional derivative along the direction d at a point (x_1, x_2, x_3) , where

$$d = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix} ?$$